

Lectures 12 & 13

In Class Exercises



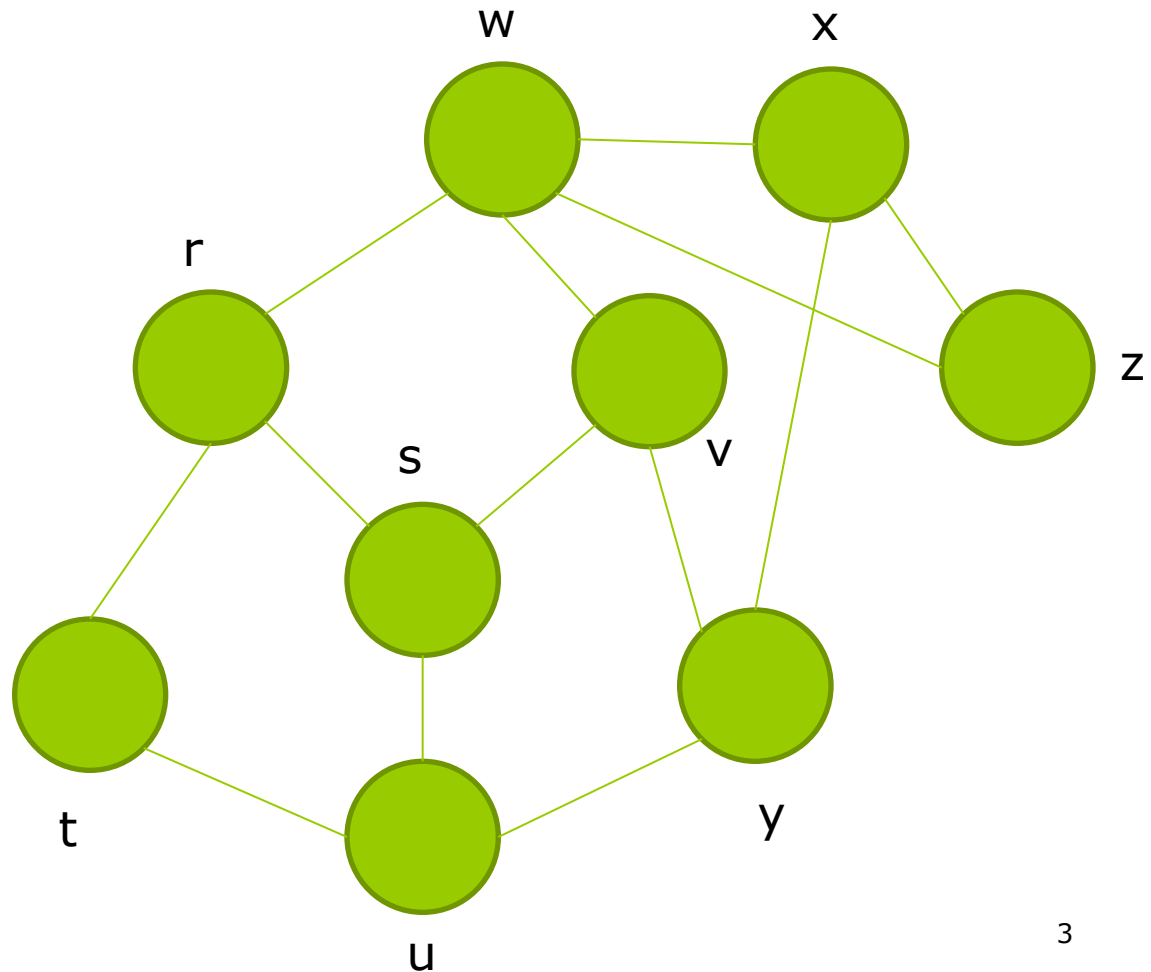
Exercise 1

What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers?

a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21

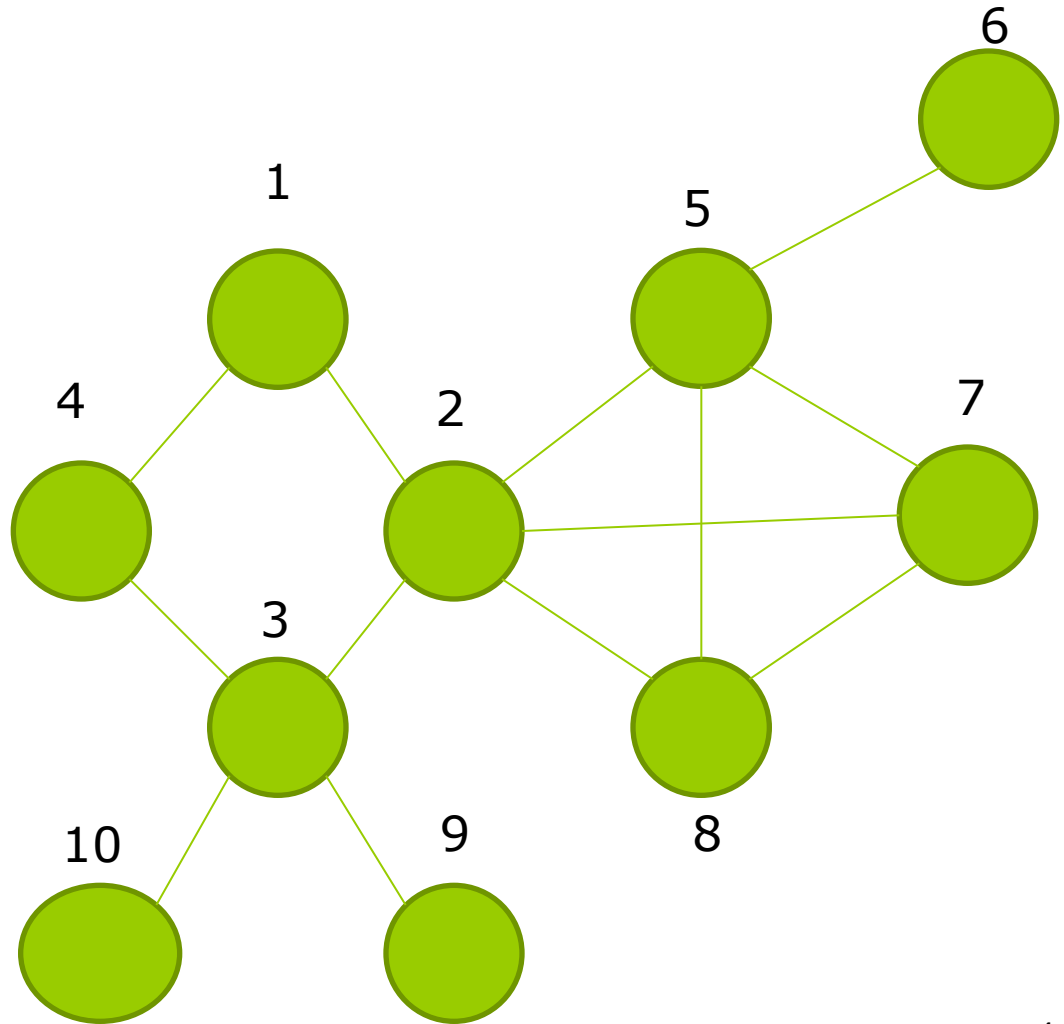
Exercise 2

Show the d (distance) and π (predecessor) values that result from running Breadth First Search on this undirected graph, using vertex **u** as the source. Assume the neighbors of a vertex are visited in alphabetical order.



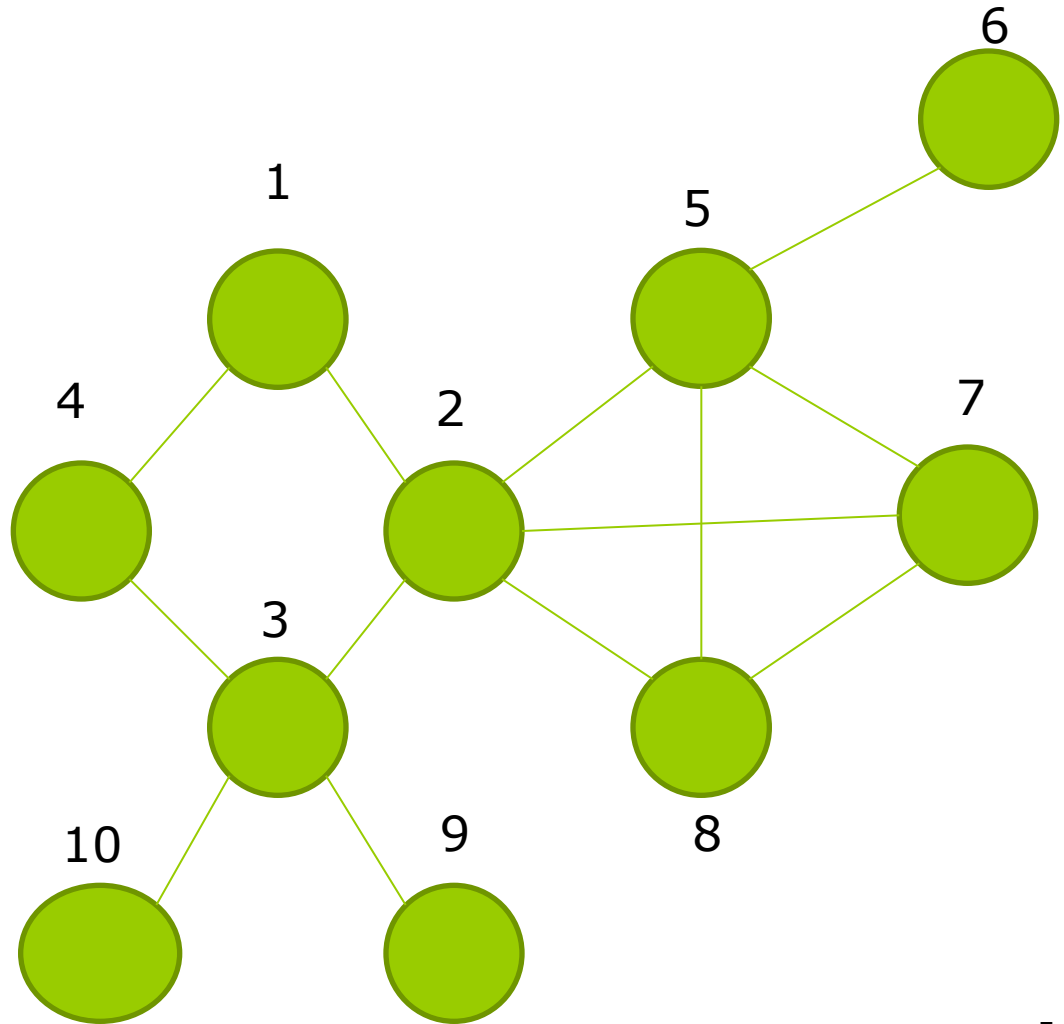
Exercise 3

Show the d (distance) and π (predecessor) values and the queue Q that result from running breadth-first search on the undirected graph of the following figure, using vertex **1** as the source.



Exercise 4

Show the d (distance) and π (predecessor) values and the queue Q that result from running breadth-first search on the undirected graph of the following figure, using vertex **5** as the source.



Exercise 5

Show how Depth first search works for the following graph. Assume that the DFS procedure considers the vertices in **alphabetical** order and assume that each adjacency list is ordered **alphabetically**. Show the **discovery** and **finish** times for each vertex.

